THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 9 & 10: **Propositions and their Converses, How Quantities Combine: Understanding Data** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The converse of the proposition “If a quadrilateral is a rhombus, then its diagonals bisect each other.” is: 1

(a) If the diagonals of a quadrilateral do not bisect each other, then it is not a rhombus.
(b) A rhombus has diagonals that bisect each other.
(c) If a quadrilateral is not a rhombus, then its diagonals do not bisect each other.
(d) If the diagonals of a quadrilateral bisect each other, then it is a rhombus.

2. Which of the following is a counterexample to the proposition “If a number is odd, then it is prime.”? 1

(a) 9 **(b)** 7 **(c)** 11 **(d)** 5

3. The proposition “If a number is a multiple of 12, then it is a multiple of 4” is true. Its converse is: 1

(a) false; 12 is not a multiple of 4
(b) true; every multiple of 4 is a multiple of 12
(c) true; 12 is a multiple of 4
(d) false; 8 is a multiple of 4 but not of 12

4. A triangle has sides 7 cm, 24 cm and 25 cm. By the converse of the Baudhāyana–Pythagoras theorem it is: 1

(a) right-angled, since all sides are different
(b) not right-angled, since the sides are not equal
(c) right-angled, since $a^2 + b^2 = c^2$
(d) right-angled, since $a + b > c$

5. The number 36 has: 1

(a) 11 factors; it is a perfect square **(b)** 9 factors; it is a perfect square
(c) 10 factors; it is not a perfect square **(d)** 11 factors; it is not a perfect square

6. A group of 15 students has an average of 82 marks and another group of 35 students has an average of 74 marks. The average of all 50 students is: 1

(a) 75.4 **(b)** 78.4 **(c)** 76.4 **(d)** 77.4

7. The averages of 3 weeks of 5 days each are 12, 15, 18. The overall average is: 1

(a) 15 **(b)** 16 **(c)** 14 **(d)** 17

8. 600 mL of a solution with 8% salt is mixed with 400 mL of a solution with 13% salt. The concentration of salt in the mixture is: 1

(a) 11% **(b)** 10% **(c)** 12% **(d)** 9%

9. 80 mL of a solution with 25% salt is mixed with 120 mL of a solution with 15% salt. The concentration of salt in the mixture is: 1

(a) 19% **(b)** 20% **(c)** 18% **(d)** 21%

10. Marks of 70, 80, 60 are combined with weights 2 : 3 : 5. The weighted average is: 1

(a) 70 **(b)** 69 **(c)** 68 **(d)** 67

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the proposition "A number has an odd number of factors when it is a perfect square" in 'If … then …' form. 2

12. How many factors does 64 have? Is the number of factors odd or even? What does this say about 64? 2

13. A cafe rates food, service and ambience with weights 4 : 3 : 2. A customer gives 5, 4 and 2. Find the weighted rating. 2

14. In a stacked bar chart, what does the total length of a bar represent? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The number $2^{2^3} + 1 = 257$ is prime. Explain why this does not prove that all numbers of the form $2^{2^n} + 1$ are prime. 3

16. Four hostel groups of 10 students each have average daily study times 2, 3, 4 and 3 hours. Find the overall average and state why simple averaging works here. 3

17. Write the following in 'If … then …' form and write its converse: "A number has exactly 3 factors when it is the square of a prime." 3

OR

A sticks-and-diagonals model: a quadrilateral is of type Q when it has equal diagonals. 3
(a) Must the two sticks have equal length?
(b) Does the way they are placed matter if the diagonals only need to be equal?

18. A shop earned ₹8000 from books at 30% profit and ₹1000 from covers at 50% profit. Find the overall profit percentage. 3

OR

A zoo has 50 deer with average weight 40 kg. The average weight of the males is 46 kg and of the females 36 kg. Form an equation for the number of males x and solve it. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Show that "If n is a product of two unequal primes, then it has exactly 4 divisors" is true using 15 and 33. 5
(ii) Is the converse true? Find a counterexample if any.

OR

Case study: Meera notices: "All the rectangles I drew had equal diagonals." 5
(i) Write this as a proposition.
(ii) Write the converse.
(iii) Draw (describe) an isosceles trapezium to show the converse is false.
(iv) Name a type of quadrilateral for which the converse is true under an extra condition.

20. (i) Make a 100% stacked bar table for the generation data (MW): 2015 — coal 600, solar 100, wind 200, hydro 100; 2025 — coal 800, solar 600, wind 400, hydro 200. 5
(ii) What do you observe about solar?

OR

A solution is made by mixing 600 mL of a 5% salt solution with 300 mL of an 8% sugar solution. 5
(i) Find the percentage of salt in the mixture.
(ii) Find the percentage of sugar.
(iii) Why are the answers not 5% and 8%?

END OF QUESTION PAPER Total: 40 Marks